

Adelic Hecke Operators — Place-Crossing Gates for Adelic Quantum Computation

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Abstract

Phase 1 of the Adelic Langlands Physics program established the local Langlands correspondence for $\mathrm{GL}(1)$ at every place of $\mathbb{Q}$ via the Silent Parameter $R(\mathrm{SU}(2))_k \cong R_v$. Phase 2 constructs the global framework. This paper defines the **adelic Hecke operators** — operators T_p acting on the restricted tensor product $\bigotimes'_v R_v$ of local character rings that have the automorphic fusion basis as eigenvectors with eigenvalues given by the local Frobenius parameters $\alpha_p(j)$. The Hecke algebra $\mathcal{H}(G(\mathbb{A})//K)$ is identified as the gate set for adelic quantum computation, where place-crossing anyon braiding implements Hecke operator composition. The eigenproperty $\mathbf{T}_p|\pi\rangle = \lambda_p(\pi)|\pi\rangle$ is the statement that the adelic anyon model diagonalizes the global Hecke algebra, connecting local Langlands data to global automorphic spectra. Three falsifiable predictions are formulated for experimental verification.

1. Introduction

The Langlands program bridges local and global phenomena: local Langlands correspondences at each place v of a number field are assembled into a global automorphic representation $\pi = \bigotimes'_v \pi_v$, whose L-function factorizes over places. The Adelic Langlands Physics program's Phase 1 established the local correspondences at every place of $\mathbb{Q}$ in terms of $\mathrm{SU}(2)_k$ fusion rings. Phase 2 now constructs the global bridge.

The central object of this paper is the **Hecke operator** T_p — the arithmetic operator at prime p that acts on automorphic forms and whose eigenvalues are the local Langlands parameters. In the adelic physics framework, T_p acquires an operational meaning: **it is a quantum gate that transforms the adelic anyon fusion state**, with eigenvalues directly encoding the Frobenius (Langlands) parameter at p .

2. Local Data (Review of Phase 1)

2.1 Local Langlands at Each Place

From Phases 1.1—1.4, we have:

Place v	Local character ring R_v	Isomorphism	Level condition
∞	$R(\mathrm{SU}(2))$ (representation ring)	Infinite level limit	$k \rightarrow \infty$
$p = 2$	$R(\mathbb{Z}_2^\times) \cong R(\mathrm{SU}(2))_{2^m-2}$	Silent Parameter	$k + 2 = 2^m$
$p = 3$	$R_m(\mathbb{Z}_3^\times) \cong R(\mathrm{SU}(2))_{2 \cdot 3^{m-1}-1}$	Silent Parameter	$k + 1 = 2 \cdot 3^{m-1}$
$p \geq 5$	$R_m(\mathbb{Z}_p^\times) \cong R(\mathrm{SU}(p-1))_k$ or higher	Phase 1.3 obstruction	$k + 1 = (p-1)p^{m-1}$

At each non-Archimedean place p , the local Langlands parameter is the Frobenius eigenvalue:

$$\alpha_p(j) = \chi_{\mathrm{tors}}(g_0) \cdot e^{2\pi i \cdot n(j)/p^{m-1}}$$

where $n(j)$ encodes the spin label modulo the conductor. [ALP Phase 1.3]

2.2 Local Fusion Rings as Hilbert Spaces

The local Hilbert space at place p with conductor m is:

$$\mathcal{H}_p(m) = \mathbb{C}[G_p(m)] \cong \mathbb{C}^{(p-1)p^{m-1}}$$

where $G_p(m) = \mathbb{Z}_p^\times / (1 + p^m \mathbb{Z}_p)$ is the finite character group. The fusion ring $R(\mathrm{SU}(2))_k$ at the level determined by p and m provides the physical state space: each simple object j corresponds to a basis vector $|j\rangle_p$ in $\mathcal{H}_p(m)$. [my conjecture]

3. The Adelic Construction

3.1 Restricted Tensor Product

The global adelic Hilbert space is the restricted tensor product of local spaces:

$$\mathcal{H}_{\mathbb{A}} = \bigotimes_{p \leq \infty}' \mathcal{H}_p$$

with respect to a choice of “unramified vector” $|0\rangle_p$ at each finite prime p . The restriction condition is: for almost all p , the state is the unramified vector — the trivial character $\chi_0 \in R_m(\mathbb{Z}_p^\times)$ corresponding to the identity character on $\mathbb{Z}_p^\times$. [established, standard adelic construction]

Physically, this means: **the adelic anyon model has trivial anyon content at all but finitely many primes** — only a finite set of “excited places” carry non-trivial topological charge.

3.2 Adelic Fusion Category

The adelic fusion category $\mathcal{F}(\mathbb{A})$ is the category whose Grothendieck ring is the restricted tensor product of local fusion rings:

$$K_0(\mathcal{F}(\mathbb{A})) = \bigotimes_{p \leq \infty}' R(\mathrm{SU}(n_p))_{k_p}$$

where for each p , the rank n_p and level k_p satisfy the Phase 1 conditions. The fusion of adelic anyons is defined componentwise:

$$(\otimes_p j_p) \otimes (\otimes_p j'_p) = \otimes_p (j_p \otimes j'_p)$$

Each local factor fuses according to the Verlinde algebra at its own level. [my conjecture]

3.3 Automorphic Representations as Basis States

An automorphic representation π of $\mathrm{GL}(1, \mathbb{A}_{\mathbb{Q}})$ decomposes as $\pi = \otimes_p' \pi_p$. Each local component π_p is a character χ_p of $\mathbb{Q}_p^\times$, corresponding under local Langlands to a simple object in the $\mathrm{SU}(2)_{k_p}$ fusion ring (Phase 1).

Therefore, the basis states of $\mathcal{H}_{\mathbb{A}}$ are labeled by:

$$|\pi\rangle = \bigotimes_p |j_p\rangle_p$$

where $j_p \in \{0, \frac{1}{2}, 1, \dots, \frac{k_p}{2}\}$ and $j_p = 0$ (trivial anyon) at almost all p . The set of all such automorphic basis states spans the global adelic Hilbert space. [my conjecture]

4. Hecke Operators

4.1 Classical Definition

For a prime p , the classical Hecke operator T_p on automorphic forms for $\mathrm{GL}(1)$ is defined by:

$$(T_p f)(x) = f(p^{-1}x) + \sum_{a=0}^{p-1} f(p^{-1}(x+a))$$

At the level of the local representation π_p , T_p acts as multiplication by the eigenvalue:

$$T_p |\pi_p\rangle = \lambda_p(\pi) |\pi_p\rangle$$

where $\lambda_p(\pi) = \pi_p(p) + \pi_p^{-1}(p)$ for $\mathrm{GL}(2)$ automorphic forms, or more simply $\lambda_p(\pi) = \pi_p(p)$ for $\mathrm{GL}(1)$. [established]

4.2 Adelic Hecke Operators

In the adelic fusion framework, the Hecke operator at p is defined as a linear operator on $\mathcal{H}_{\mathbb{A}}$:

$$\mathbf{T}_p : \mathcal{H}_{\mathbb{A}} \longrightarrow \mathcal{H}_{\mathbb{A}}$$

that acts non-trivially only on the p -adic factor:

$$\mathbf{T}_p = \text{id}_{\infty} \otimes \left(\bigotimes_{q \neq p} \text{id}_q \right) \otimes T_p \otimes \left(\bigotimes_{q \neq p} \text{id}_q \right)$$

On the p -adic factor $\mathcal{H}_p(m)$, the local Hecke operator T_p is defined by the fusion matrix:

$$(T_p)_{j,j'} = \langle j' | \mathbf{T}_p | j \rangle = \alpha_p(j) \cdot \delta_{j,j'}$$

That is, T_p is diagonal in the fusion basis with eigenvalues $\alpha_p(j)$ — the Frobenius eigenvalue from the local Langlands correspondence (Phase 1). This is the eigenproperty.

Physically: the Hecke operator at p measures the p -adic anyon charge — braiding the adelic anyon around the p -adic Bruhat—Tits tree returns the phase $\alpha_p(j)$. [speculative]

4.3 Hecke Algebra as Gate Set

The full Hecke algebra $\mathcal{H}(G(\mathbb{A})//K)$ is generated by $\{\mathbf{T}_p\}_{p \text{ prime}}$ subject to the commutation relations:

$$[\mathbf{T}_p, \mathbf{T}_q] = 0 \quad \text{for } p \neq q$$

since Hecke operators at different primes act on different local factors. The algebra is a commutative $\mathbb{C}$ -algebra spanned by the simultaneous eigenvectors $|\pi\rangle$:

$$\mathbf{T}_p |\pi\rangle = \alpha_p(\pi) |\pi\rangle$$

This commutativity is the statement that **adelic anyon braiding operations at different primes commute** — the Hecke gates form a mutually commuting set of quantum operations, analogous to the stabilizer group in topological quantum computation. [my conjecture]

5. Physical Interpretation

5.1 Hecke Gates as Quantum Operations

Each Hecke operator $\mathbf{T}_p$ is a **unitary quantum gate** on the adelic Hilbert space (unitarity follows from $|\alpha_p(j)| = 1$ at unramified places). The gate set is:

$$\mathcal{G}_{\text{Hecke}} = \{\mathbf{T}_p : p \text{ prime}\}$$

The action of $\mathbf{T}_p$ on an adelic state $|\pi\rangle$ is:

1. **Read out the p -adic anyon label j_p** from the local factor
2. **Multiply by the Frobenius phase $\alpha_p(j_p)$**
3. **Leave all other local factors unchanged**

This is a **local gate** in the adelic sense — it acts only at a single place. [my conjecture]

5.2 Gate Complexity

The physical cost of implementing $\mathbf{T}_p$ grows with the conductor m . The gate time τ_p is proportional to the size of the Bruhat—Tits tree branch at depth m :

$$\tau_p \propto p^{m-1}$$

reflecting that the Hecke operator must resolve the p^{m-1} distinct anyon types at conductor m . For $p = 2$, this is $\tau_2 \propto 2^{m-1}$; for $p = 3$, $\tau_3 \propto 3^{m-1}$. [speculative]

5.3 Adelic Anyon Braiding

The composition of Hecke operators $\mathbf{T}_{p_1} \mathbf{T}_{p_2}$ corresponds to **simultaneous anyon braiding at two different primes**. Since the operators commute, the order of braiding does not matter — a feature unique to the commutative Hecke algebra for $\mathrm{GL}(1)$.

For $\mathrm{GL}(n)$ with $n > 1$, the Hecke algebra is non-commutative, and the composition encodes non-abelian anyon statistics — a richer class of adelic topological quantum gates. [speculative]

6. Global L-Function

The global L-function associated to an automorphic representation $\pi = \otimes_p \pi_p$ factorizes over places:

$$\Lambda(s, \pi) = L_\infty(s, \pi_\infty) \cdot \prod_p L_p(s, \pi_p)$$

where each local factor is:

$$L_p(s, \pi_p) = \frac{1}{1 - \alpha_p(j) \cdot p^{-s}}$$

and $\alpha_p(j)$ is the eigenvalue of $\mathbf{T}_p$ on $|\pi\rangle$ — the Hecke eigenvalue.

The completed L-function $\Lambda(s) = N^{s/2} \cdot \pi^{-s/2} \Gamma(s/2) \cdot \prod_p L_p(s)$ satisfies the functional equation $\Lambda(1-s) = \varepsilon \cdot \Lambda(s)$, where $\varepsilon = \prod_p \alpha_p(0)$ is the product of Hecke eigenvalues at the trivial character. [established for $\mathrm{GL}(1)$]

Physical interpretation: The global L-function is the **partition function of the adelic anyon model**, with the local factors $L_p(s)$ as the partition sums at each prime. [speculative, to be developed in Phase 2.2]

7. Falsifiable Predictions

1. **[P1] Hecke eigenproperty.** For the adelic fusion basis $|j\rangle = \otimes_p |j_p\rangle_p$, the Hecke operator $\mathbf{T}_p$ acts diagonally:

$$\mathbf{T}_p |j\rangle = \alpha_p(j_p) \cdot |j\rangle$$

with $\alpha_p(j_p) = (-1)^{2j_p} \cdot e^{2\pi i(2j_p)/p^{m-1}}$ for SU(2) levels. [my conjecture] - **Falsification:** Compute the Verlinde fusion matrix at level $k = (p-1)p^{m-1} - 1$ and verify the eigenproperty directly. - **[CHECK: 2027-06]** Numerical diagonalization of fusion matrices for $p = 2, 3, 5$ at $m = 1, 2, 3$.

2. **[P2] Hecke algebra commutativity.** $[\mathbf{T}_p, \mathbf{T}_q] = 0$ for all p, q . [my conjecture]
 - **Falsification:** Find a counterexample in the adelic tensor product where two local Hecke operators fail to commute.
 - **[CHECK: 2027-12]** Formal proof or counterexample.
3. **[P3] Gate complexity scaling.** The physical gate time for $\mathbf{T}_p$ at conductor m scales as p^{m-1} . [speculative]
 - **Falsification:** Measure or simulate gate times for small (p, m) .
 - **[CHECK: 2028-12]** Requires Phase 3/4 lattice simulations.

8. Conclusion

The adelic Hecke operators $\mathbf{T}_p$ provide the bridge from local Langlands data (Phase 1) to global automorphic spectra. Each $\mathbf{T}_p$ is a quantum gate acting on the restricted tensor product $\bigotimes'_v R_v$ of local fusion rings, diagonal in the automorphic fusion basis with eigenvalues $\alpha_p(j_p)$. The Hecke algebra generated by $\{\mathbf{T}_p\}$ is the gate set for adelic quantum computation, where anyon braiding at different primes implements Hecke operator composition.

The global L-function factorizes over places with each local factor determined by the Hecke eigenvalue, connecting the algebraic (Langlands) and physical (anyonic) descriptions. Phase 2.2 will formalize the L-function as a physical observable on adelic Hilbert space, and Phase 2.3 will connect Langlands functoriality to renormalization group flow.

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